EDUSOUL - An Integrated Explainable AI Platform for Academic Guidance, Careers, and Wellbeing

R26-IT-073

The Problem: The "Lost" Post-A/L Student

- **Information Overload & Fragmentation:** Students struggle to navigate complex UGC Z-score cutoffs, district quotas, and evolving industry demands simultaneously.
- **The Competency Mismatch:** There is a systemic failure in the education-to-employment pipeline where graduates possess academic qualifications but lack the specific, functional technical skills required by the modern job market.
- **The Productivity-Wellbeing Gap:** High academic pressure leads to burnout, yet students lack data-driven tools to balance study loads with mental health and lifestyle constraints.
- **The Trust Deficit:** Online career advice is often unverified; there is a lack of secure, authenticated mentorship connecting students with proven industry professionals.



R26-IT-073

USERS & STAKEHOLDERS

Who is Affected ?

- **Primary Users:** Post-G.C.E. A/L students , undergraduates, and Degree holders in Sri Lanka.
- **Secondary Users:** Professionals
- **Stakeholders:** * **Educational Institutions:** To improve student retention and success rates.
- **Industry/Employers:** Access to a workforce with skills aligned to SLASSCOM 2026 standards.
- **Mentors/Alumni:** A structured, secure platform to give back to the student community.

IMPACT OF THE PROBLEM

To the User: Wrong degree → academic failure / dropout

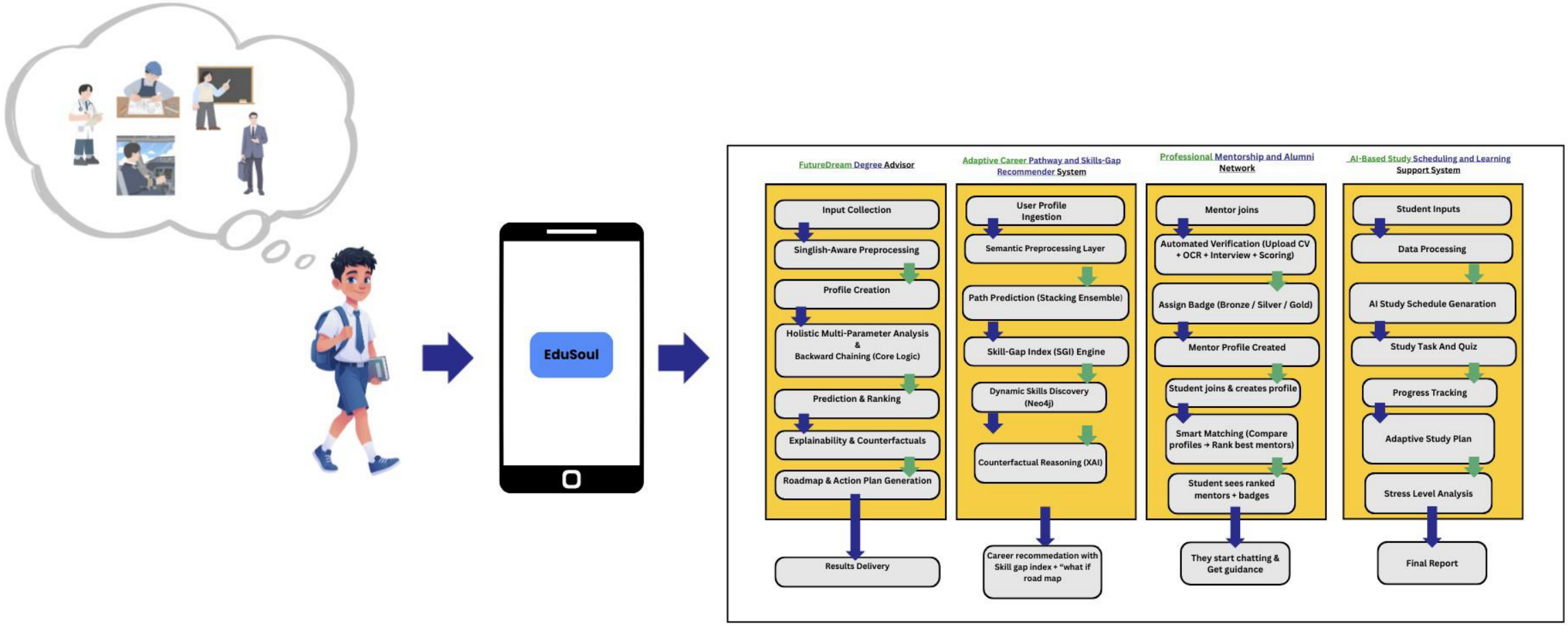
- Mental stress, low confidence, family pressure
- Delayed career start → years lost

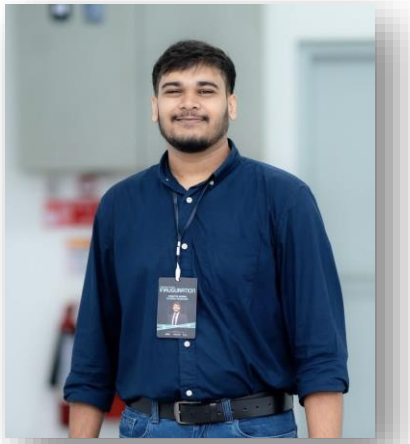
To Society: High graduate unemployment / underemployment

- Brain drain (students go abroad for better fit)
- National loss: wasted free education + talent mismatch

EXISTING SOLUTIONS & THEIR LIMITATIONS

- **UGC Handbooks/Websites:** Static and data-heavy; they provide "what" is available but not "how" it fits a student's specific personality or lifestyle.
- **Generic Career Coaching:** Often subjective and lacks "Backward-Chaining" logic or real-time job market data.
- **Standard To-Do List Apps:** They don't account for *stress levels* or *burnout prediction* – they treat students like machines, not humans.
- **Unverified LinkedIn/Social Groups:** High risk of misinformation and lack of structured, verified mentorship "trust-chains."


EduSoul
FutureDream Degree Advisor
Input Collection
Singlish-Aware Preprocessing
Profile Creation
Holistic Multi-Parameter Analysis & Backward Chaining (Core Logic)
Prediction & Ranking
Explainability & Counterfactuals
Roadmap & Action Plan Generation
Results Delivery
Adaptive Career Pathway and Skills-Gap Recommender System
User Profile Ingestion
Semantic Preprocessing Layer
Path Prediction (Stacking Ensemble)
Skill-Gap Index (SGI) Engine
Dynamic Skills Discovery (Neo4j)
Counterfactual Reasoning (XAI)
Career recommedation with Skill gap index + "what if road map
Professional Mentorship and Alumni Network
Mentor joins
Automated Verification (Upload CV + OCR + Interview + Scoring)
Assign Badge (Bronze / Silver / Gold)
Mentor Profile Created
Student joins & creates profile
Smart Matching (Compare profiles → Rank best mentors)
Student sees ranked mentors + badges
They start chatting & Get guidance
AI-Based Study Scheduling and Learning Support System
Student Inputs
Data Processing
AI Study Schedule Genaration
Study Task And Quiz
Progress Tracking
Adaptive Study Plan
Stress Level Analysis
Final Report



IT22220288 | RUSIRU L.W.S

FutureDream Degree Advisor

Problem Definition:

* **The Scale:** 340,000+ A/L students face extreme pressure annually with limited guidance.

* **The Mismatch:** Over-reliance on Z-scores ignores travel tolerance, stress, and family attachment.

* **The Crisis:** High dropout rates and **19% youth unemployment** (2025) due to degree-career mismatch.

The Research Gap:

* **Linear Bias:** Current systems only look "Forward" from grades, ignoring the "Backward" dream goal.

* **Data Silos:** Lack of Multifactor Analysis (Personality + Lifestyle + Market Sustainability).

* **Black-Box Models:** No Explainable AI (XAI) or "What-if" (Counterfactual) guidance.

* **Language Barrier:** No Singlish-aware trilingual mobile solutions for natural interaction.

Proposed Solution (The System):

Backward Career Reasoning: Starts with the "Dream Job" and maps the path back to the degree.

Holistic Predictor: Integrates Grades + Big Five Personality + Entrepreneurial Mindset + Lifestyle.

Sustainable Roadmapping: Generates a full path: Bachelor's → Masters → Skills → Milestones.

Trilingual Interface: Singlish-aware NLP for Sinhala, Tamil, and English natural interaction.

Novelty & Innovation:

First-in-Class: First Sri Lankan system to use backward-chaining for post-A/L career intelligence.

XAI & Counterfactuals: Provides "If-Then" guidance (e.g., *"If you improve your English, your chance for X career rises by 20%"*).

Local Contextualization: Factors in District Quotas, Hostel Life, and Travel Tolerance.

Ready Datasets: Need to make Data sets for A/L performance , Market Trends.

FUTUREDREAM DEGREE ADVISOR

Existing Work/tool/product	Features available	Limitations
CareerMe.lk (National Career Guidance Platform – Sri Lanka)	Aptitude & interest tests, basic degree/course suggestions, trilingual UI (Sinhala/Tamil/English), career interest mapping, some job suggestions	Forward-only (current grades → suggestion), no backward reasoning from dream career goal, no personality/lifestyle/emotional factors, no XAI explanations, no counterfactual guidance, no long-term roadmap
UGC / University Admission Portals (e.g., admission.ugc.ac.lk + counselling)	Z-score cutoff lookup, basic stream/degree eligibility advice	Purely cutoff-based, no personalization, no dream goal input, no XAI, no lifestyle factors
LinkedIn Career Explorer + Learning Recommendations (global tool, widely used in Sri Lanka)	job role suggestions, course/degree recommendations, interest-based career paths	Not localized (no Z-score/district quotas, no Sri Lankan admission data), forward-only, no pre-university focus, black-box recommendations, no lifestyle compatibility, no Singlish/trilingual support

FutureDream Degree Advisor

Research Methodology:

Knowledge Domain: Explainable Artificial Intelligence (XAI) in Personalized Educational Decision Support

•**Collect data** → Dream career goal + A/L results + personality + lifestyle + interests (public datasets + synthetic data)

•**Backward reasoning** → Start from dream goal → find required degree & skills

•**Holistic analysis** → Score academic + personality + lifestyle + job market factors together

•**Prediction**

• **Explainability**

• **Counterfactuals**

• **Roadmap**

•**Mobile UI**

•**Backend Test & evaluate**

Latest Technologies

AI & Data Dependencies

- **XGBoost** - XGBoost predicts the success probability for each degree
- **SHAP -** explains why- for example, 'Your analytical thinking contributed 22%'
- **DiCE** - shows 'what-if' advice – like 'If you improve Physics to B, chance rises 35%'
- pandas + NumPy → data preprocessing & multi-parameter scoring
- Supabase

Common / Shared Components (across platform)

- Singlish-aware input preprocessor
- XAI utilities (SHAP/DiCE)

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Validation & Evaluation Strategy:

- **Model performance**: 10-fold cross-validation → target ≥85% accuracy
- **Explanation quality**: Human evaluation (Likert scale) on clarity & usefulness (20–30 students)
- **Counterfactual usefulness**: Survey on practicality of "if-then" advice
- **Usability**
- **Overall satisfaction**
- **Mobile testing**

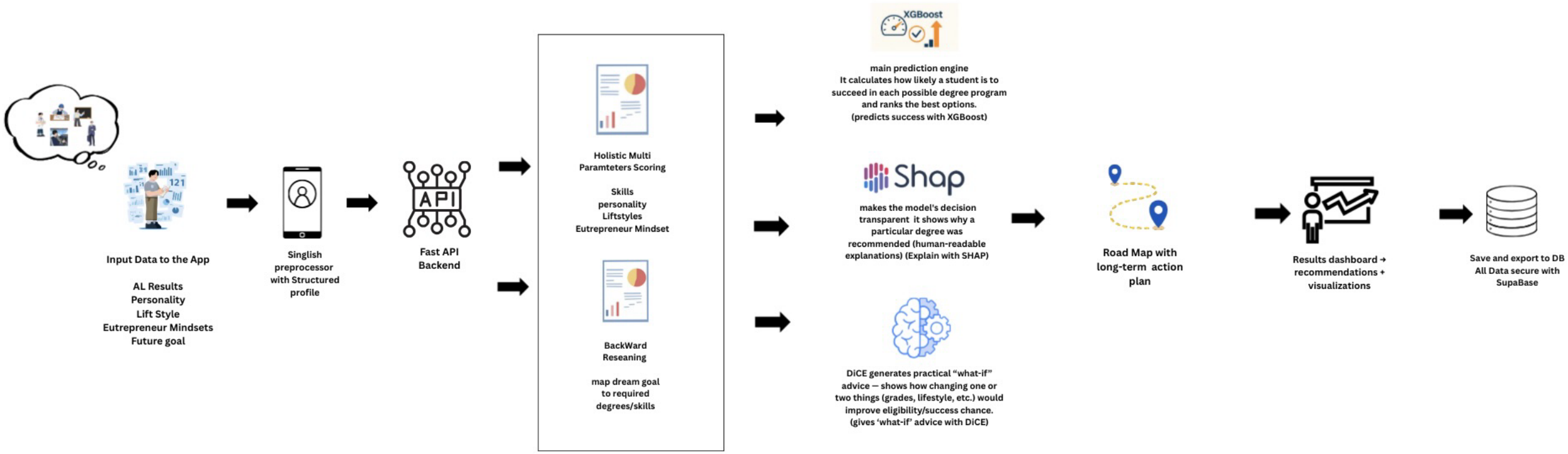
Data & Ethics:

- **Data availability**: Need to make Data sets + synthetic generation (Faker/NumPy)
- **Security**: HTTPS + JWT authentication, anonymized storage, no raw academic records kept
- **Ethical clearance**: university ethics approval for any future user testing
- **Privacy**: GDPR principles + university guidelines followed

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System Overview:



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IT22152114 | WICKREMATHILAKA J.A

ADAPTIVE CAREER PATHWAY AND SKILLS-GAP RECOMMENDER SYSTEM

Problem Definition :

- **Absence of Quantifiable Gap Measurement:** Existing portals fail to quantify the specific "semantic distance" between a student and a target role.

- **Underprepared Graduates**: Traditional systems suggest high-value roles based on a degree title, ignoring that local students may lack the specific, applied competencies missing in the current workforce.

Research Objectives :

Main Objective: To design an **Explainable AI (XAI)-driven system** that utilizes a **Stacking Ensemble** to identify career paths and a **Skill-Gap Index (SGI)** to mathematically quantify professional competency gaps for actionable, prescriptive guidance.

Sub-Objectives:

- **Dynamic Skills Ontology:** Employs a Knowledge Graph (Neo4j) to infer hidden talents by mapping relational skill adjacencies rather than relying on static keyword matching.
- **High-Precision Predictive Modelling:** To build a Stacking Ensemble (Random Forest, SVM, and DNN) achieving a predictive accuracy of **90%** in identifying optimal career paths.
- **Evaluation and Bias Auditing:** Actionability Score and perform fairness auditing mechanisms.

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ADAPTIVE CAREER PATHWAY AND SKILLS-GAP RECOMMENDER SYSTEM

Existing Work/tool/product	Features available	Limitations
Recruitment Portals (e.g., Rooster.jobs, TopJobs.lk)	Rooster.jobs, TopJobs.lk)Real-time vacancy listing, automated shortlisting (ATS), and interview scheduling.	**Recruitment-Centric**: Focused on employers' needs; lacks personalized, trilingual skills-gap roadmaps for candidates.
Global AI Discovery Tools (e.g., CareerExplorer)	Personalized role matching based on interests, skills, and personality assessments.	**Contextual Void**: Lacks localization for the Sri Lankan market
Industry Portals (e.g., SLASSCOM Careers)	Industry-wide skills analysis reports and competency persona definitions.	**Static & General**: only Provides broad industry standards.

ADAPTIVE CAREER PATHWAY AND SKILLS-GAP RECOMMENDER SYSTEM

Research Methodology:

- **Design Science Research (DSR)** iterate through the Relevance (identifying local gaps), Design (building the AI), and Evaluation cycles to solve real-world career guidance problems.

- **Intelligence Pipeline:** Preprocessing ➞ Ontology Mapping ➞ Path Prediction ➞ SGI Calculation ➞ XAI Generation ➞Optimization.

- **Data Strategy:** Ingests **10,000+ localized job descriptions** (via BeautifulSoup/Selenium) and **SLASSCOM 2026 personas** utilizing **spaCy-driven NLP** to transform these raw industry standards into mathematical "Target Vectors" (*I*).

Latest Technologies Utilized :

- **Predictive Layer:** Stacking Ensemble integrating **Random Forest**, **SVM**, and **Deep Neural Networks (DNN)** for 90%+ classification accuracy.

- **Semantic Layer: spaCy NLP** for Named Entity Recognition (NER) and **Neo4j** for a graph-based Skills Ontology mapping skill dependencies.

- **Interpretability Layer: SHAP** and **Counterfactual Reasoning** modules used to generate human-readable "What-If" career roadmaps.

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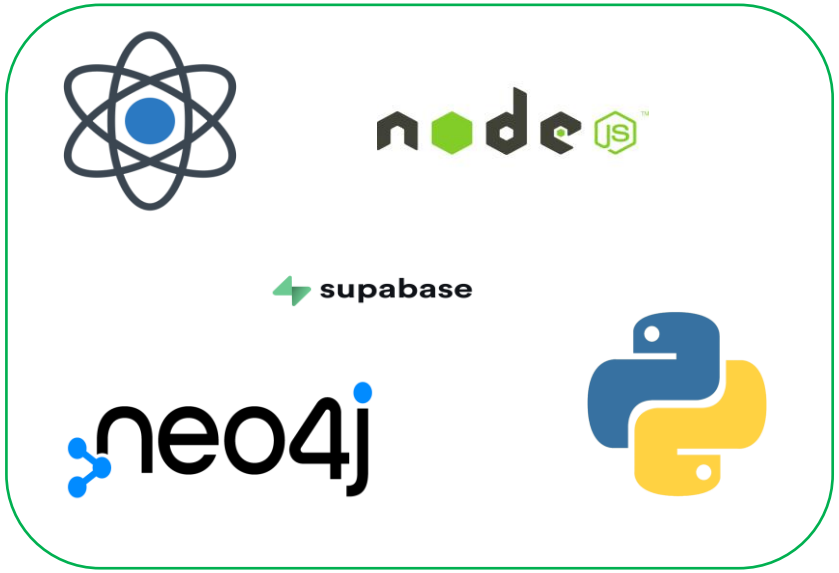
ADAPTIVE CAREER PATHWAY AND SKILLS-GAP RECOMMENDER SYSTEM

Validation, Security & Ethics

- **Validation:** Evaluated via **Actionability Scores** , Utilize **Mean Absolute Error (MAE)** to validate the exactness of the **(SGI).**
- **Security:** Stateless processing and **JWT authentication** ensure compliance with the **Sri Lankan PDPA No. 9 of 2022**.
- **Ethical & Bias Auditing: Fairness Metric:** The **Statistical Parity Difference (SPD)** Output will remain unbiased across different age groups, genders, and regional backgrounds.

Technical Dependencies :

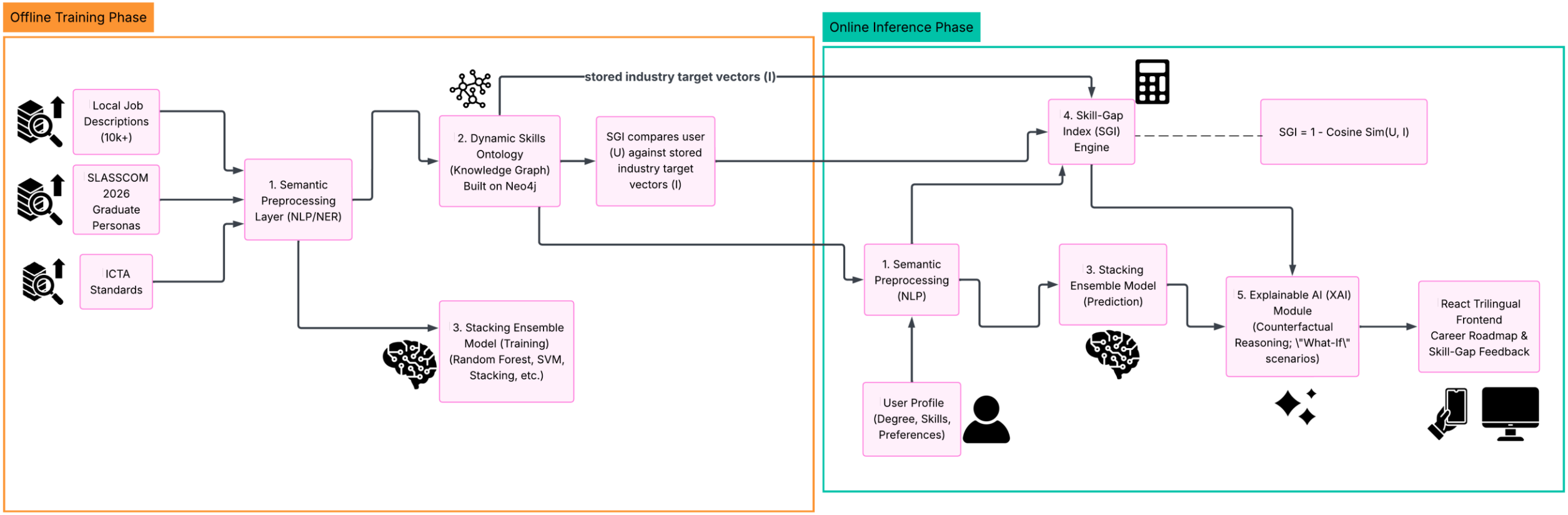
- **Backend Orchestration:** Requires **Node.js/Express** to facilitate asynchronous communication between the Python analytical microservice and the frontend.
- **Data Persistence:** Relies on **Supabase** for storing unstructured user profile data and **Neo4j** for graph relationships.
- **Common Integrated Components:** * **Trilingual Interface:** A shared **React.js** frontend supporting English, Sinhala, and Tamil for inclusive accessibility.



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ADAPTIVE CAREER PATHWAY AND SKILLS-GAP RECOMMENDER SYSTEM

System Overview:



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IT22150370 | MAHIKALPA K.B.V.I

PROFESSIONAL MENTORSHIP AND ALUMNI NETWORK

1. The Problem & Gap

- **Problem:** Lack of trusted mentors and weak verification leading to privacy risks.
- **Research Gap:** Existing systems rely on slow manual validation and lack privacy-preserving tech.

2. Proposed Solution

- **The Network:** A secure platform connecting students with verified university & industry mentors.
- **Matching:** Intelligent pairing based on degree programs and career goals.

3. The Technology (Tasks)

- **OCR Verification:** Automated CV/Certificate scanning.
- **AI Interview:** Credibility scoring system.
- **Secure Chat:** Real-time professional communication.

4. Novelty (What's New?)

- **Multi-layered Trust-Chain:** Automated doc verification + interview scoring.
- **Privacy-Preserving:** No permanent storage of sensitive files.
- **Badge System:** Instant credibility via Bronze/Silver/Gold levels.

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PROFESSIONAL MENTORSHIP AND ALUMNI NETWORK

Existing Work/tool/product	Features available	Limitations
ScholarX (Sri Lanka)	6-month free mentorship for selected students Matching with Sri Lankan diaspora experts Focus on academic & career goals	Only for selected few (not open to all) Fixed 6 months only Manual matching, no auto verification or badges No real-time chat or privacy-focused document checks
MentorcliQ (Global)	Advanced skill/goal-based matching Profiles, messaging & chat Goal tracking & analytics Used in some university programs	Expensive (enterprise pricing) Not focused on Sri Lankan students No auto OCR/interview verification or badges Limited local context & trilingual support
University Alumni Networks (e.g., NIBM, Moratuwa, Colombo)	Alumni mentor connections Career advice from same university Free for students/alumni Events & networking	Only one university at a time Mostly manual connections No auto matching, badges or verification No real-time chat or national platform

PROFESSIONAL MENTORSHIP AND ALUMNI NETWORK

Research Methodology

- **Knowledge Domain:** Trustworthy Artificial Intelligence (TAI) and Intelligent Student Support Systems
- **Core domain:** Automated mentor verification + intelligent mentor–student matching in educational guidance systems
- **Contributes:** Transparent and trustworthy mentorship support through multi-layered verification, badge-based credibility scoring, and privacy-preserving data handling.
- **Educational domain**
- **Verification domain**
- **Privacy and security domain**
- **Communication domain**
- **Sri Lankan context**
- **Novel methodological direction**

Latest Technologies





OCR
Optical Character Recognition.

supabase



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PROFESSIONAL MENTORSHIP AND ALUMNI NETWORK

Validation & Evaluation Strategy

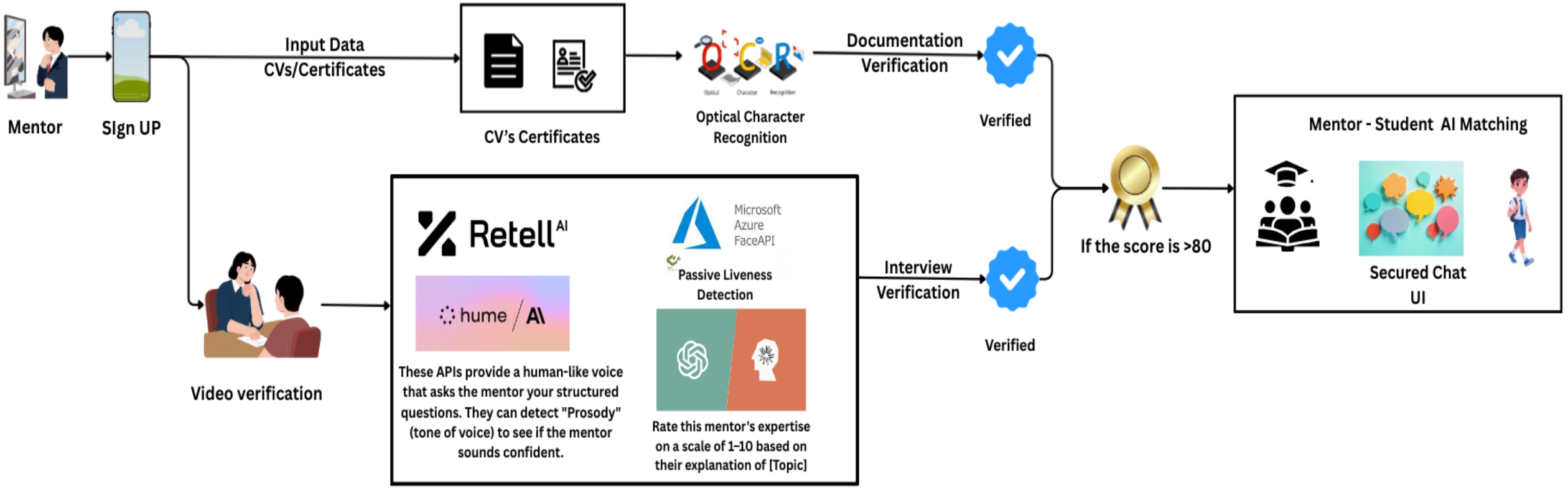
- **Mentor Verification Accuracy:** 10-fold cross-validation → target ≥85% (Precision, Recall, F1-score)
- **Matching Quality:** Human evaluation (20–30 students) on relevance of recommended mentors
- **Trust Badge Clarity:** Likert scale survey on usefulness of Bronze/Silver/Gold badges
- **Counterfactual Guidance:** Survey on practicality of "if-then" mentor suggestions
- **Usability:** SUS target ≥80; load time <3s; offline caching; smooth chat experience
- **Compatibility:** Android 8.0+ & iOS 12+
- **Overall Satisfaction:** NPS + open user feedback

Data & Ethics

- **Data Sources:** Public datasets (alumni records, academic profiles, job market trends, Big Five personality) + synthetic data (Faker / NumPy)
- **Security:** HTTPS, JWT authentication, encrypted temporary storage, raw documents deleted after processing
- **Ethical Clearance:** Prototype uses no primary data; future testing requires university ethics approval
- **Privacy:** GDPR + university guidelines followed; metadata-only extraction, no permanent storage of sensitive documents

PROFESSIONAL MENTORSHIP AND ALUMNI NETWORK

System Overview:



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PROFESSIONAL MENTORSHIP AND ALUMNI NETWORK

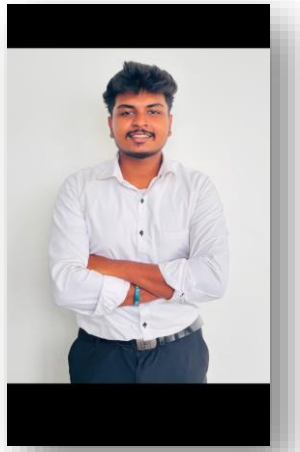
Dependencies & Common Components:

Frontend Dependencies
- React Native + Expo → Cross-platform app (iOS & Android)
- Redux / Context API → State management

Backend Dependencies
- Node.js + Express.js → REST APIs for verification, matching, chat
- Socket.IO → Real-time mentor–student chat
- MongoDB / Firebase / Supabase → User profiles, mentor data, messages
- JWT / Firebase Auth → Secure authentication

AI & Data Dependencies
- PaddleOCR / TrOCR / LayoutLM → Document verification
- Sentence Transformers / BERT → Scoring questionnaires, mentor–student matching
- pgvector / Qdrant → Vector search for intelligent matching
- SHAP / DiCE → Explainable AI + counterfactual guidance
- pandas + NumPy → Data preprocessing and scoring

Common / Shared Components (across platform)
- Supabase / Firebase → Shared database across modules
- XAI utilities → Reusable explanation modules (SHAP/DiCE)
- Badge & Trust System → Bronze/Silver/Gold verification badges
- Input Preprocessor → Standardized multilingual text input



IT22150370 | WITHANAGE .D.K

AI-Based Study Scheduling and Learning Support System

1. Problem Definition .

- Students manage **lectures, assignments, exams, and part-time jobs**.
- Difficult to **manage time and study schedules**.
- Leads to **missed deadlines, low productivity, and stress**.

2. Proposed Solution

- Develop an **AI-based study assistant system**.
- Students upload **timetable, work hours, exam modules, Sleeping hours and lecture slides**.
- System **generates daily study tasks and quizzes** to support exam preparation.

3. The Research Gap

- Existing apps only provide **reminders and simple task lists**.
- They **do not generate study schedules using the student timetable**.
- They **do not evaluate learning progress or understanding**.

4. Novelty Highlights

- **AI automatic study scheduling** based on timetable and workload.
- **Lecture-based task generation** for exam preparation.
- **Quiz-based learning evaluation** to analyze student performance.

AI-BASED STUDY SCHEDULING AND LEARNING SUPPORT SYSTEM

Existing Work/tool/product	Features available	Limitations
My Study Life (Student Planner App)	Manage timetable, assignments, exams, reminders, and daily schedule in one app.	Only basic scheduling and reminders. No AI-based learning analysis or adaptive study planning.
Notion Study Planner	Custom study planners with tasks, calendars, notes, and progress tracking in one workspace.	Expensive (enterprise pricing) Not focused on Sri Lankan students.
Motion AI Scheduling App	AI automatically schedules tasks in free time based on deadlines and calendar	Focuses on task scheduling only; not designed specifically for student learning evaluation or exam preparation.

AI-Based Study Scheduling and Learning Support System

Research Methodology

- Requirement Analysis
- Data Collection
- Data Preprocessing
- Machine Learning Model Development
- Task Scheduling Algorithm
- Lecture Material Analysis
- Quiz Generation
- Performance Analysis
- System Testing and Evaluation

Latest Technologies

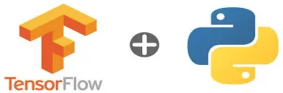








OCR
Optical Character Recognition.









AI-Based Study Scheduling and Learning Support System

Validation & Evaluation Strategy

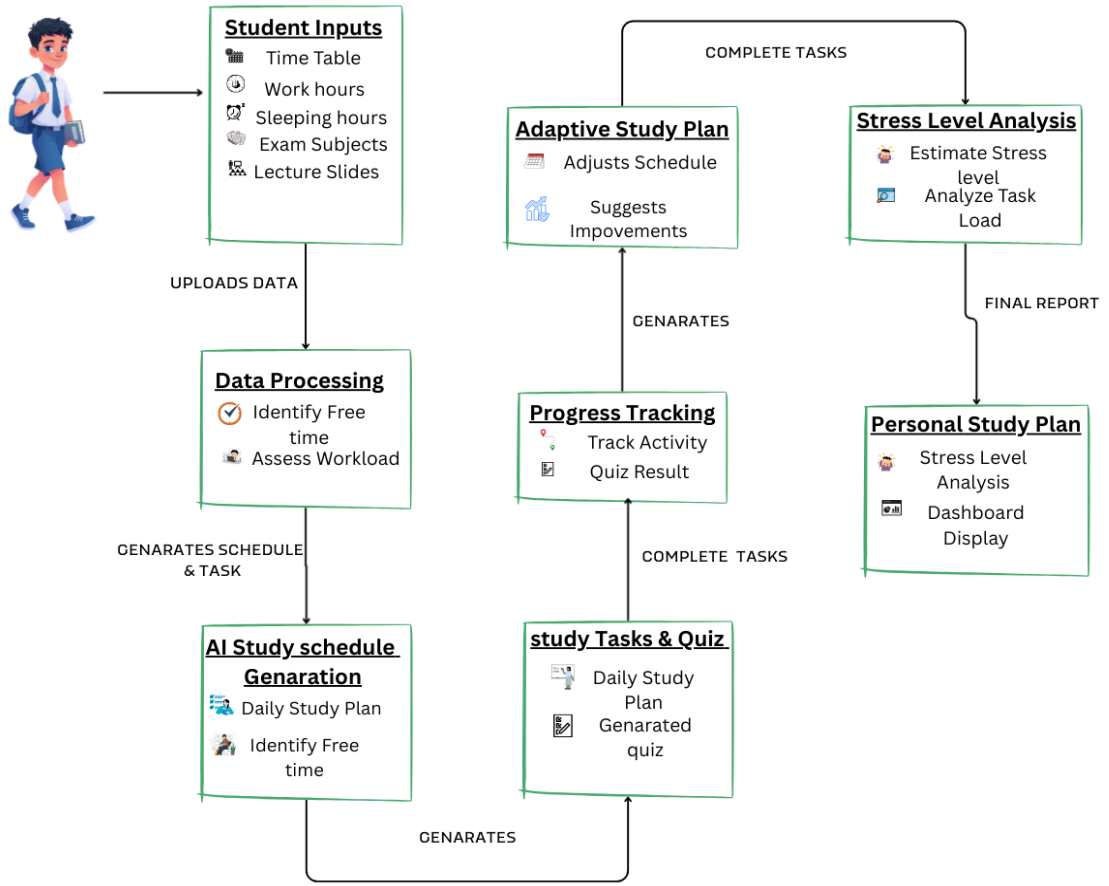
- **System Testing** – Test whether the system correctly generates study schedules and tasks.
- **Model Accuracy Evaluation** – Measure the accuracy of the machine learning model used for analysis.
- **User Testing** – Allow students to use the system and collect their feedback.
- **Quiz Performance Analysis** – Evaluate student quiz results to check learning improvement.
- **Usability Evaluation** – Check if the system is easy for students to use.
- **Performance Analysis** – Analyze whether the system helps improve student productivity and study management

Data & Ethics

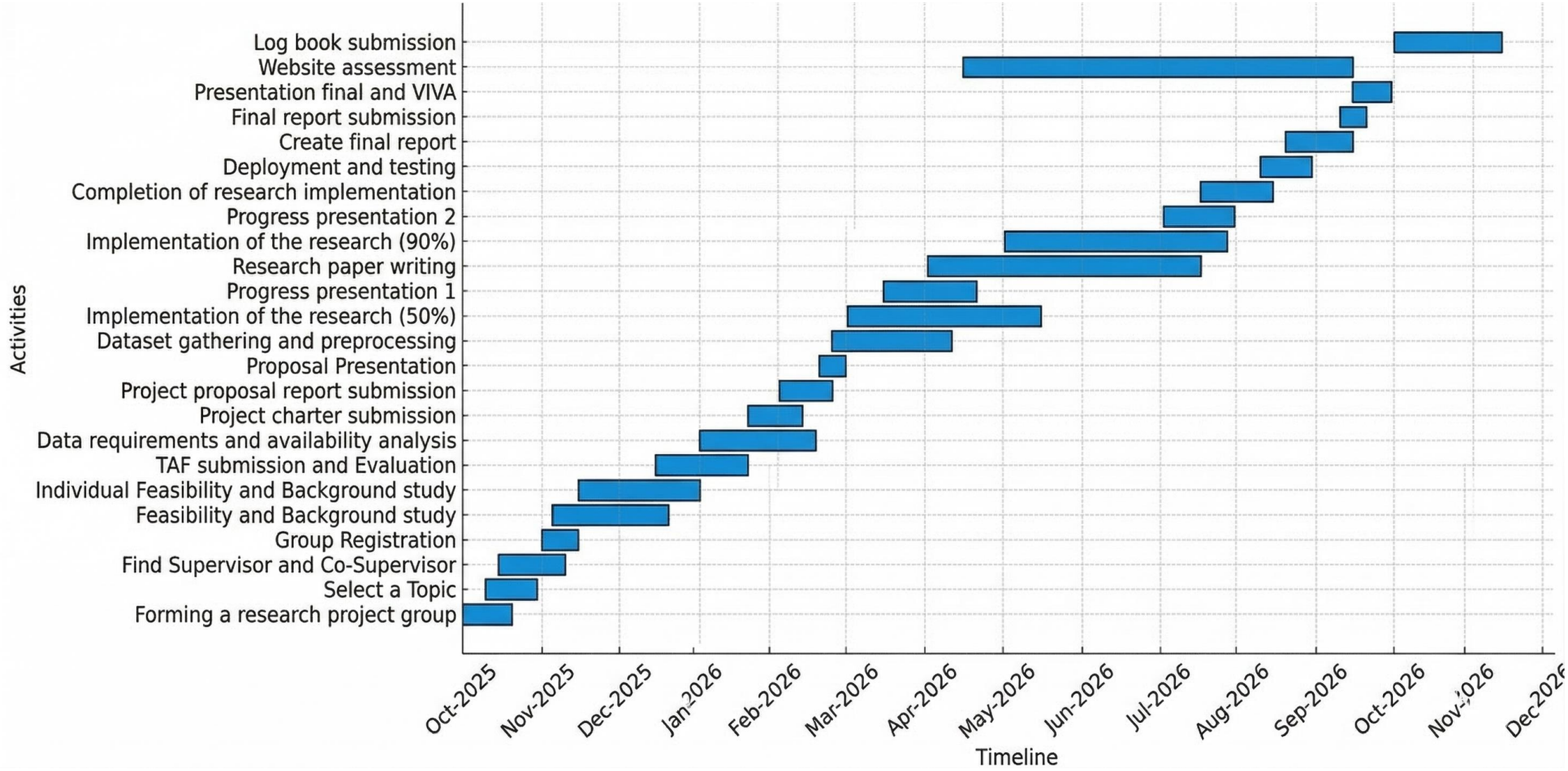
- **Data Sources:** Public datasets (academic stress level maintance, Student productivity distraction, student mental health) + synthetic data(Faker / NumPy)
- **User Consent** – Students will give permission before their data is collected or used.
- **Secure Data Storage** – All collected data will be stored securely in the system database.
- **Ethical Data Use** – The collected data will only be used for research and improving the system.

AI-Based Study Scheduling and Learning Support System

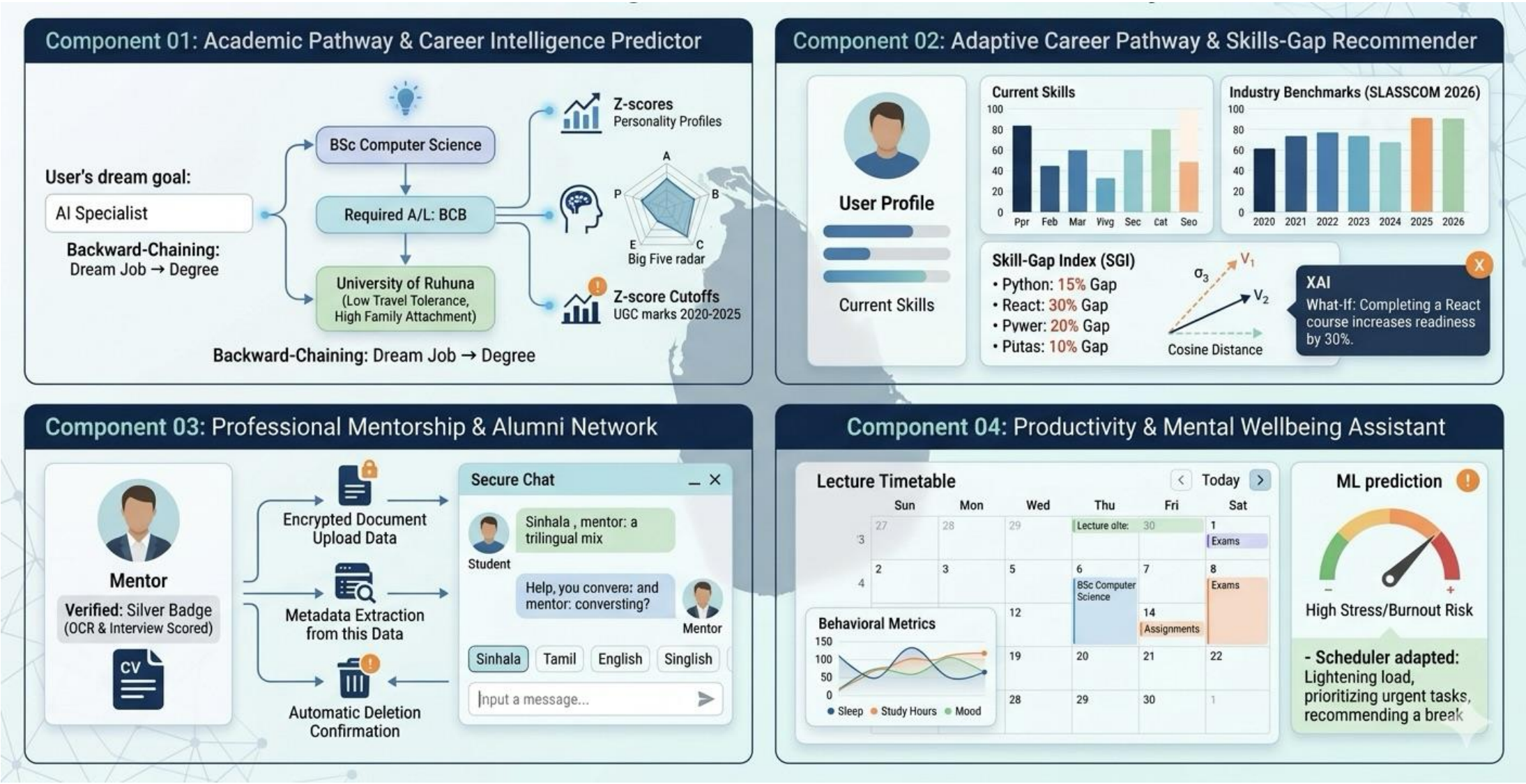
System Overview:



Gantt chart


Log book submission
Website assessment
Presentation final and VIVA
Final report submission
Create final report
Deployment and testing
Completion of research implementation
Progress presentation 2
Implementation of the research (90%)
Research paper writing
Progress presentation 1
Implementation of the research (50%)
Dataset gathering and preprocessing
Proposal Presentation
Project proposal report submission
Project charter submission
Data requirements and availability analysis
TAF submission and Evaluation
Individual Feasibility and Background study
Feasibility and Background study
Group Registration
Find Supervisor and Co-Supervisor
Select a Topic
Forming a research project group
Activities
Oct-2025
Nov-2025
Dec-2025
Jan-2026
Feb-2026
Mar-2026
Apr-2026
May-2026
Jun-2026
Jul-2026
Aug-2026
Sep-2026
Oct-2026
Timeline

Work Breakdown Structure



Thank you !

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